

KnowServe Corp. — Engineering Department

DevOps Deployment Pipeline

Overview

KnowServe's CI/CD pipeline automates build, test, and deployment across staging and production.

Average deployment frequency: 4 per week.

Stages

- 1 . Build – Docker images built via GitHub Actions; artifacts stored in ECR .
- 2 . Test – Unit + integration tests (pytest) run in parallel; minimum 90 % success rate required .
3. Security Scan – SAST and dependency checks block critical findings .
- 4 . Deploy – Automated rollout to Kubernetes cluster using ArgoCD with blue/green strategy .
5. Monitor – Prometheus + Grafana collect metrics; alerts trigger via Slack if CPU > 80 % for 5 min.

Change Management

All production deployments require sign-off from DevOps Lead + QA Manager.

Rollback window = 30 min; automated recovery scripts restore last stable image.

KPIs

- Mean Time to Deploy (MTTD) < 20 min
- Mean Time to Recovery (MTTR) < 15 min
- Failed Deployment Rate < 2%

Documentation

Pipeline diagrams and YAML templates stored in ``/infrastructure/docs/pipeline.``